

PROBLEM SOLVING AND TESTING USING JAVA

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WEEK-3 PROBLEMS

1. Java Date and Time – <https://www.hackerrank.com/challenges/java-date-and-time/problem>

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;

class Result {

    /*
     * Complete the 'findDay' function below.
     *
     * The function is expected to return a STRING.
     * The function accepts following parameters:
     * 1. INTEGER month
     * 2. INTEGER day
     * 3. INTEGER year
     */
}
```

```

public static String findDay(int month, int day, int year) {
    // Using java.time.LocalDate to easily find the day of the week
    java.time.LocalDate dt = java.time.LocalDate.of(year, month,
day);

    // Returns the day of the week in capital letters (e.g.,
WEDNESDAY)
    return dt.getDayOfWeek().toString();
}

}

public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new
InputStreamReader(System.in));

        BufferedWriter bufferedWriter = new BufferedWriter(new
FileWriter(System.getenv("OUTPUT_PATH")));

        String[] firstMultipleInput =
bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

        int month = Integer.parseInt(firstMultipleInput[0]);

        int day = Integer.parseInt(firstMultipleInput[1]);

```

```
int year = Integer.parseInt(firstMultipleInput[2]);

String res = Result.findDay(month, day, year);

bufferedWriter.write(res);
bufferedWriter.newLine();

bufferedReader.close();
bufferedWriter.close();
}
}
```

Congratulations

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

[Next Challenge](#)

✓ Test case 0	Compiler Message	
✓ Test case 1 🔒	Success	
✓ Test case 2 🔒	Input (stdin)	Download
✓ Test case 3 🔒		
✓ Test case 4 🔒	Expected Output	Download
✓ Test case 5 🔒		

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1

WEDNESDAY

2. Number of Days Between Two Dates – <https://leetcode.com/problems/number-of-days-between-two-dates/>

```
import java.time.LocalDate;
```

```
import java.time.temporal.ChronoUnit;
```

```
class Solution {
```

```
    public int daysBetweenDates(String date1, String date2) {
```

```
        LocalDate d1 = LocalDate.parse(date1);
```

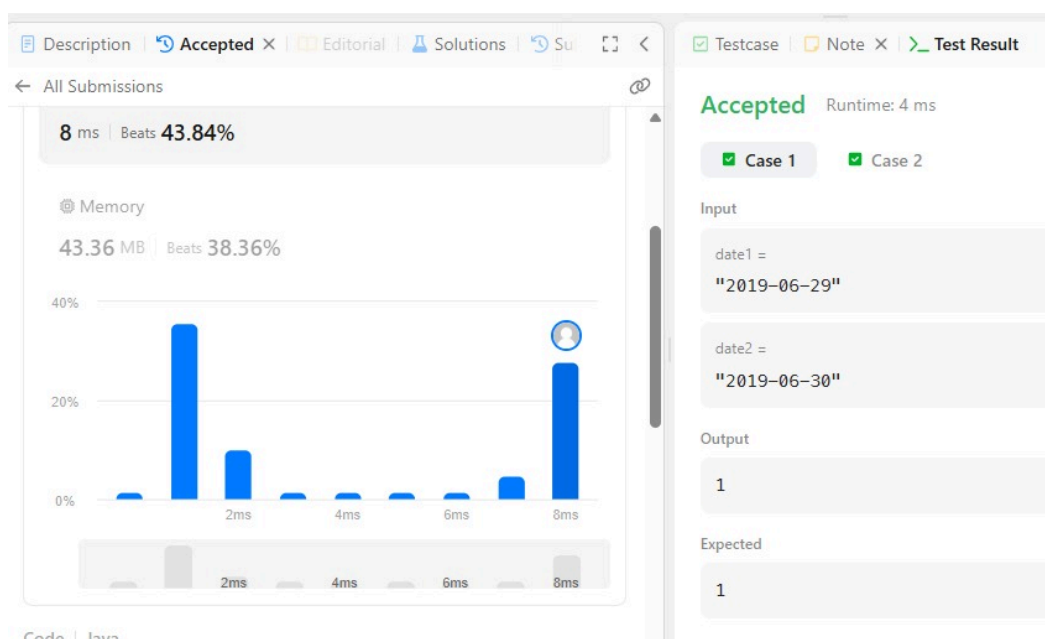
```
        LocalDate d2 = LocalDate.parse(date2);
```

```
        // Use ChronoUnit.DAYS.between to find the absolute difference  
        in days
```

```
        return (int) Math.abs(ChronoUnit.DAYS.between(d1, d2));
```

```
    }
```

```
}
```



3.Day of the Year – <https://leetcode.com/problems/day-of-the-year/>

```
class Solution {  
    public int dayOfYear(String date) {  
        // Parse year, month, and day from the string  
        int year = Integer.parseInt(date.substring(0, 4));  
        int month = Integer.parseInt(date.substring(5, 7));  
        int day = Integer.parseInt(date.substring(8, 10));  
  
        // Days in each month for a non-leap year  
        int[] daysInMonths = { 31, 28, 31, 30, 31, 30, 31, 31, 30,  
31, 30, 31 };  
  
        // Check for leap year and update February's days if  
        necessary  
        if (isLeapYear(year)) {  
            daysInMonths[1] = 29;  
        }  
  
        // Sum up the days of the preceding months plus the  
        current day  
        int dayOfYear = day;
```

```

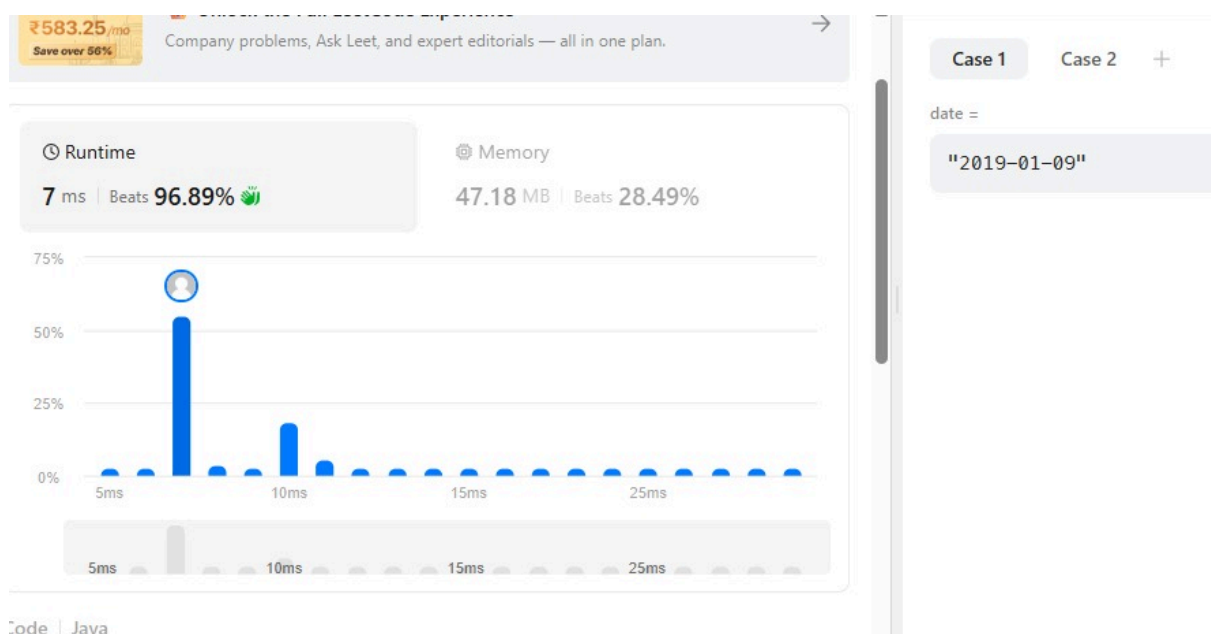
    for (int i = 0; i < month - 1; i++) {
        dayOfYear += daysInMonths[i];
    }

    return dayOfYear;
}

private boolean isLeapYear(int year) {
    // A year is a leap year if it is divisible by 4 and not 100,
    // or if it is divisible by 400.

    return (year % 4 == 0 && year % 100 != 0) || (year %
400 == 0);
}
}

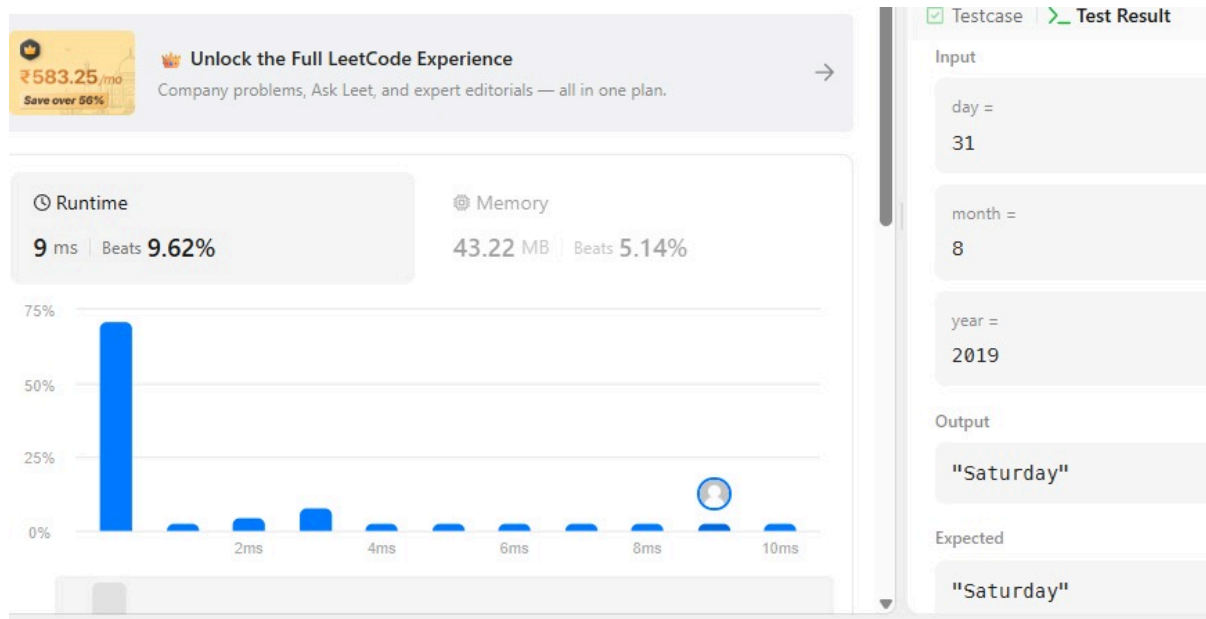
```



4. Day of the Week – <https://leetcode.com/problems/day-of-the-week/>

```
import java.time.LocalDate;
import java.time.format.TextStyle;
import java.util.Locale;

class Solution {
    public String dayOfTheWeek(int day, int month, int year) {
        LocalDate date = LocalDate.of(year, month, day);
        return date.getDayOfWeek().getDisplayName(TextStyle.FULL,
        Locale.ENGLISH);
    }
}
```



5. Java Priority Queue – <https://www.hackerrank.com/challenges/java-priority-queue/problem>

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
import java.util.Comparator;
import java.util.PriorityQueue;

class Student {
    private int id;
    private String name;
    private double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }
}
```

```
public String getName() {  
    return name;  
}
```

```
public double getCGPA() {  
    return cgpa;  
}  
}
```

```
class Priorities {  
    public List<Student> getStudents(List<String> events) {  
        PriorityQueue<Student> pq = new PriorityQueue<>(  
            Comparator.comparing(Student::getCGPA).reversed()  
                .thenComparing(Student::getName)  
                .thenComparing(Student::getID)  
        );  
  
        for (String event : events) {  
            String[] parts = event.split("\\s+");  
            if (parts[0].equals("ENTER")) {  
                String name = parts[1];  
                double cgpa = Double.parseDouble(parts[2]);  
                int id = Integer.parseInt(parts[3]);  
                pq.add(new Student(id, name, cgpa));  
            }  
        }  
    }  
}
```

```

        } else if (parts[0].equals("SERVED")) {
            pq.poll();
        }
    }

    List<Student> students = new ArrayList<>();
    while (!pq.isEmpty()) {
        students.add(pq.poll());
    }

    return students;
}

}

public class Solution {
    private final static Scanner scan = new Scanner(System.in);
    private final static Priorities priorities = new Priorities();

    public static void main(String[] args) {
        int totalEvents = Integer.parseInt(scan.nextLine());
        List<String> events = new ArrayList<>();

        while (totalEvents-- != 0) {
            String event = scan.nextLine();

```

```

        events.add(event);
    }

    List<Student> students = priorities.getStudents(events);

    if (students.isEmpty()) {
        System.out.println("EMPTY");
    } else {
        for (Student st: students) {
            System.out.println(st.getName());
        }
    }
}
}
}
}
}

```

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

✓ Test case 0

✓ Test case 1

✓ Test case 2

✓ Test case 3

✓ Test case 4

✓ Test case 5

✓ Test case 6

Success

Input (stdin)

1 12
2 ENTER John 3.75 50
3 ENTER Mark 3.8 24
4 ENTER Shafaet 3.7 35
5 SERVED
6 SERVED
7 ENTER Samiha 3.85 36
8 SERVED
9 ENTER Ashley 3.9 42
10 ENTER Maria 3.6 46

6. Java ArrayList – <https://www.hackerrank.com/challenges/java-arraylist/problem>

```
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);

        // Read the number of lines
        int n = scan.nextInt();

        // Create an ArrayList of ArrayLists to store the integers for each
        line
        ArrayList<ArrayList<Integer>> lines = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            int d = scan.nextInt();
            ArrayList<Integer> row = new ArrayList<>();
            for (int j = 0; j < d; j++) {
                row.add(scan.nextInt());
            }
            lines.add(row);
        }
    }
}
```

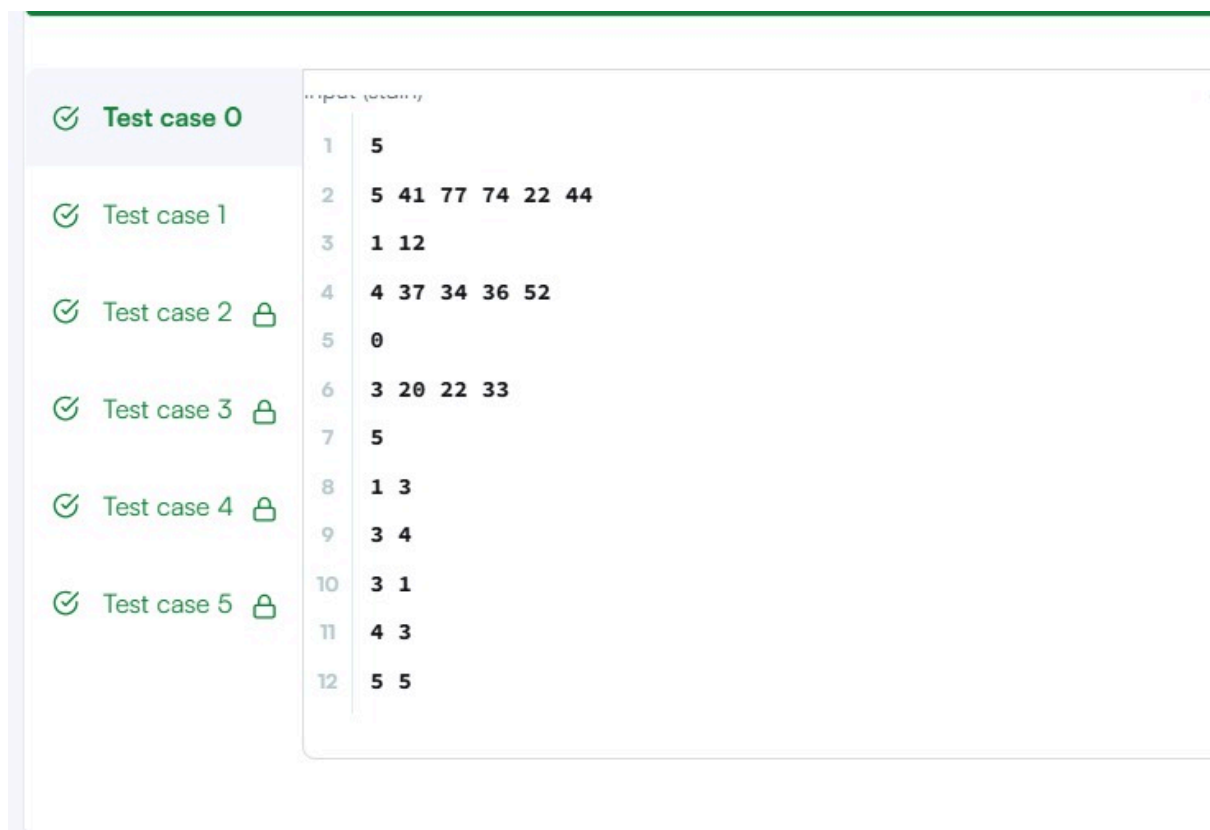
```
}

// Read the number of queries
int q = scan.nextInt();

for (int i = 0; i < q; i++) {
    int x = scan.nextInt(); // Line number
    int y = scan.nextInt(); // Position number

    // Convert 1-based indexing to 0-based indexing
    try {
        System.out.println(lines.get(x - 1).get(y - 1));
    } catch (IndexOutOfBoundsException e) {
        System.out.println("ERROR!");
    }
}

scan.close();
}
}
```



7. Largest Number – <https://leetcode.com/problems/largest-number/>

```
import java.util.Arrays;
```

```
class Solution {  
    public String largestNumber(int[] nums) {  
        // Convert the integer array to a String array  
        String[] strNums = new String[nums.length];  
        for (int i = 0; i < nums.length; i++) {  
            strNums[i] = String.valueOf(nums[i]);  
        }  
    }  
}
```

```
// Sort the strings using a custom comparator

// Compare (b + a) with (a + b) to decide the descending order
arrangement

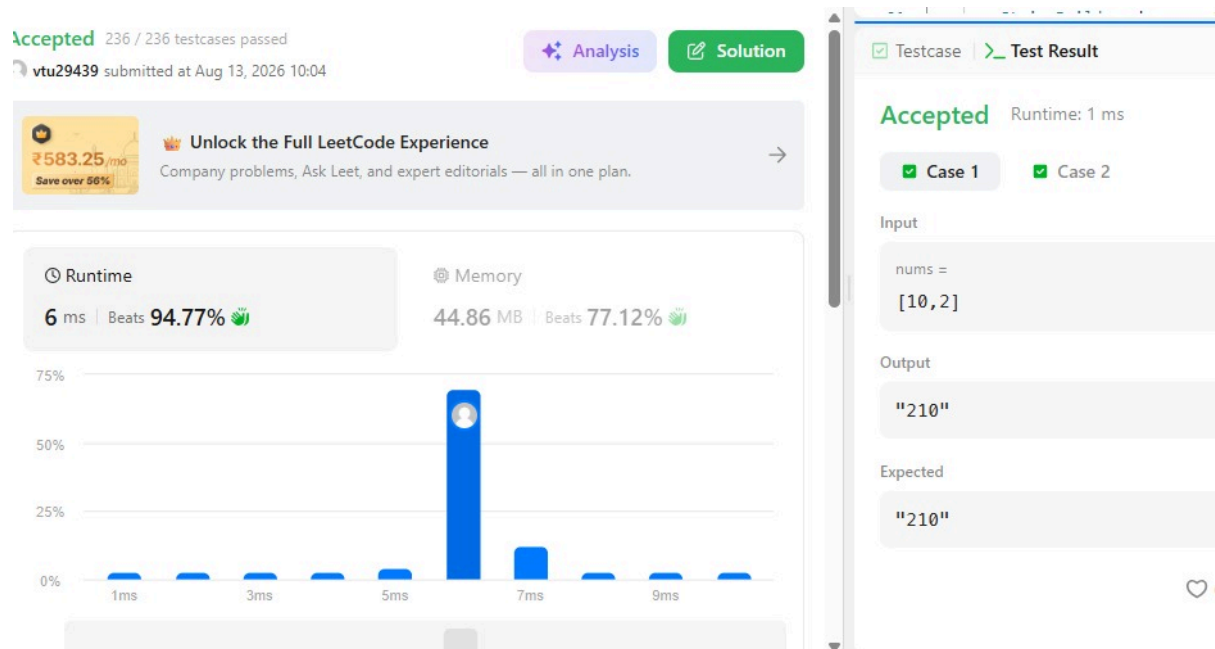
Arrays.sort(strNums, (a, b) -> (b + a).compareTo(a + b));

// Edge case: if the highest number is "0", the entire array
consists of zeros

if (strNums[0].equals("0")) {
    return "0";
}

// Build the final largest number string
StringBuilder sb = new StringBuilder();
for (String str : strNums) {
    sb.append(str);
}

return sb.toString();
}
}
```



8. Java Comparator: <https://www.hackerrank.com/challenges/java-comparator/problem>

```
import java.util.*;

class Checker implements Comparator<Player> {

    @Override
    public int compare(Player a, Player b) {
        if (a.score != b.score) {
            return Integer.compare(b.score, a.score); // Decreasing order
            of score
        }
        return a.name.compareTo(b.name); // Alphabetical order of name
    }
}

class Player{
```

```
String name;
```

```
int score;
```

```
Player(String name, int score){
```

```
    this.name = name;
```

```
    this.score = score;
```

```
}
```

```
}
```

```
class Solution {
```

```
    public static void main(String[] args) {
```

```
        Scanner scan = new Scanner(System.in);
```

```
        int n = scan.nextInt();
```

```
        Player[] player = new Player[n];
```

```
        Checker checker = new Checker();
```

```
        for(int i = 0; i < n; i++){
```

```
            player[i] = new Player(scan.next(), scan.nextInt());
```

```
        }
```

```
        scan.close();
```

```
        Arrays.sort(player, checker);
```

```

for(int i = 0; i < player.length; i++){
    System.out.printf("%s %s\n", player[i].name, player[i].score);
}
}
}

```

The screenshot shows a coding platform interface with a list of test cases on the left and a detailed view of the input and expected output on the right.

Test Cases:

- Test case 0 (Passed)
- Test case 1 (Passed)
- Test case 2 (Passed)
- Test case 3 (Passed)
- Test case 4 (Passed)
- Test case 5 (Passed)

Input (stdin):

```

1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150

```

Expected Output:

```

1 aleksa 150
2 amy 100

```

9.Sort the People- <https://leetcode.com/problems/sort-the-people/>

```
import java.util.Arrays;
```

```

class Solution {
    public String[] sortPeople(String[] names, int[] heights) {
        int n = names.length;

```

```
// Create an array of indices to keep track of the original positions
```

```
Integer[] indices = new Integer[n];
```

```
for (int i = 0; i < n; i++) {
```

```
    indices[i] = i;
```

```
}
```

```
// Sort the indices based on the heights in descending order
```

```
Arrays.sort(indices, (a, b) -> Integer.compare(heights[b], heights[a]));
```

```
// Build the result array using the sorted indices
```

```
String[] result = new String[n];
```

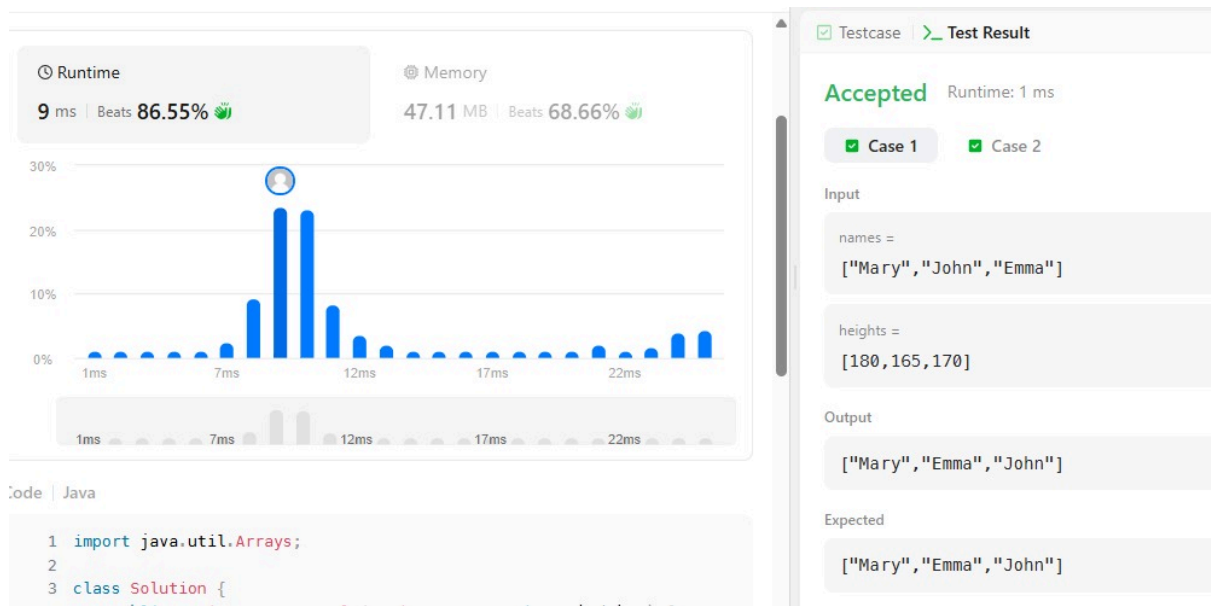
```
for (int i = 0; i < n; i++) {
```

```
    result[i] = names[indices[i]];
}
```

```
return result;
```

```
}
```

```
}
```



THANK YOU